

Literature closure for Swanepoel’s infinite-dimensional equilateral-set problem

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Source problem

In the final remarks of Konrad J. Swanepoel’s expository paper *Equilateral sets in finite-dimensional normed spaces* (arXiv:math/0406264), the subsection on infinite dimensions asks:

Does there exist an infinite equilateral set in any separable infinite-dimensional normed space?

Equivalently, the question asks whether every separable infinite-dimensional normed space contains an infinite subset whose pairwise distances are all equal.

Later answer

E. Glakousakis and S. Mercourakis, *Examples of infinite dimensional Banach spaces without infinite equilateral sets* (arXiv:1502.02500), give a negative answer. Their abstract states that they construct an infinite-dimensional separable Banach space, not isomorphic to a subspace of ℓ_1 , with no infinite equilateral sets. More explicitly, their Theorem 2.2 proves that a large class of ℓ_1 -sums of finite-dimensional spaces can be renormed so as to contain no equilateral sequence, and the final corollary takes $X_n = \ell_p^n$ for $p > 2$ to obtain a separable Banach space with an unconditional basis, not isomorphic to a subspace of ℓ_1 , and with no equilateral sequence.

Claim. *The problem from arXiv:math/0406264 has a negative literature answer.*

Proof. Let X be the Banach space obtained in the final corollary of Glakousakis–Mercourakis. Then X is separable and infinite dimensional. A Banach space is, in particular, a normed space. The same corollary says that X does not contain an equilateral sequence; equivalently, it has no countably infinite equilateral set. Any infinite equilateral subset of a metric space contains a countably infinite equilateral subset by choosing a sequence of distinct points. Thus X contains no infinite equilateral set. This is a counterexample to the universal assertion that every separable infinite-dimensional normed space contains an infinite equilateral set. $\square$

Scope note

This packet only closes the infinite-dimensional final-remarks problem. It does not resolve the finite-dimensional questions in Swanepoel’s paper, such as Kusner’s questions for ℓ_p^n , Petty’s example, the strictly convex upper-bound problems, or the conjecture that every n -dimensional normed space has an equilateral set of size at least $n + 1$.

References

- [1] K. J. Swanepoel, *Equilateral sets in finite-dimensional normed spaces*, arXiv:math/0406264, 2004.
- [2] E. Glakousakis and S. Mercourakis, *Examples of infinite dimensional Banach spaces without infinite equilateral sets*, arXiv:1502.02500, 2015.